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Multimodal glaucoma classification using segmentation based biomarker extraction

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ABSTRACT

Glaucoma is a progressive eye disease that leads to irreversible vision loss and, if not addressed promptly, can result in blindness. While various treatment options such as eye drops, oral medications, and surgical interventions exist, the disease may still progress. Therefore, early detection and diagnosis can mitigate the devastating vision loss of glaucoma. The primary focus of the proposed study is the development of a robust pipeline for the efficient extraction of quantifiable data from fundus images. What sets our approach apart from other approaches is the emphasis on automated feature extraction of glaucoma biomarkers from particular regions of interest (ROI) along with the utilization of a weighted average of image features and clinical measurements. Our unique approach leverages segmentation based models in conjunction with computer vision techniques to efficiently extract and compute clinical glaucoma biomarkers such as optic disc area and optic cup to disc ratio. The extracted biomarkers are consistent with trends observed in clinical practice, thus supporting the validity of the feature extraction approach. While subtle disease progression, inconsistencies in image quality and a general lack of metadata impact classification performance, the weighted combination of image based features with glaucoma biomarkers achieves a test accuracy of 92.63% for glaucoma classification, successfully addressing the limitations of traditional single-modality approaches such as fundus imaging and optical coherence tomography (OCT).

1. Introduction

Glaucoma is an eye disease that causes damage to the optic nerve resulting in vision loss in the affected eye and in severe cases, even blindness. According to Schellack et al. [1] glaucoma is mainly treated via lowering the intraocular pressure (IOP), which has shown to minimize damage to the optic nerve. Meedeniya et al. [2] provide a comprehensive review of deep learning approaches for glaucoma classification from retinal fundus images, including methods based on CNNs, transformers, unsupervised learning, and large language models. However, many existing studies focus primarily on unimodal approaches and are often evaluated on a limited number of datasets, which may restrict their generalizability. To address these limitations, the proposed study employs a hybrid multimodal framework that integrates complementary data modalities for glaucoma classification. Furthermore, the approach is evaluated on the SMDG-19 dataset, to improve scalability, thereby addressing the limitations associated with unimodal models and single-dataset evaluations. This multimodal strategy enables the model to leverage both structural information from images and quantitative clinical biomarkers, resulting in a more comprehensive representation of glaucoma-related changes and improved classification performance. Coan et al. [3] review methods for automatic detection of glaucoma from fundus images and highlight the versatility of artificial intelligence based techniques such as image segmentation and classification. Serener and Serte [4] utilize an ensemble of CNNs for glaucoma classification. This allows them to leverage the strengths of different models for the classification task, however, due to a lack of model diversity the GoogleLeNet [5] and ResNet50 [6] ensemble achieves an accuracy of only 77%.

In the early stages, glaucoma may present with normal IOP and often does not cause any symptoms. However, in the later stages, patients may experience blurred vision, tunnel vision and blind spots. Accurate detection and diagnosis of glaucoma is essential in preventing progressive vision loss and mitigating the effects of the disease. There is no cure for glaucoma, however, early detection and timely treatment can help prevent progression of the disease and assist in preventing blindness among patients.

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The proposed study introduces a hybrid multimodal approach for classification of glaucoma from retinal fundus images incorporating 2 different data modalities. Crucially, this study addresses the research question of how different data modalities influence glaucoma classification and evaluates the effectiveness of a multimodal classification framework. Since structural and functional changes in glaucoma do not always occur simultaneously, relying on a single data modality may delay diagnosis. Image data provides rich structural and contextual information about ocular anatomy, whereas numeric clinical data provides objective, quantifiable measurements of disease progression. Integrating these complementary sources, structural imaging and quantitative clinical biomarkers, enables a more concrete assessment of glaucoma, improving diagnostic accuracy and supporting earlier and more reliable detection. U-Net [7] based image segmentation along with custom image processing techniques help extract fundus measurements from retinal images. Concatenation of numeric and image based features in the form of a weighted average allows the multimodal model to harness clinical parameters along with structural information extracted from the fundus images using 3 vision transformers [8–10]. The proposed work advances methodological knowledge in multimodal deep learning for glaucoma classification by explicitly integrating quantitative anatomical measurements with rich visual features. Statistical analysis demonstrates the importance of these clinical biomarkers in providing complementary information, enabling the multimodal model to outperform single-modality approaches. Importantly, the framework is validated across heterogeneous multi-dataset settings with variable image quality, highlighting both its robustness and methodological contribution. The proposed multimodal approach outperforms existing approaches that consider only single data modalities to predict glaucoma from retinal fundus images in heterogeneous multi-dataset settings with variable image quality. The key contributions of this research can be articulated as follows:

- Extracting glaucoma biomarkers as quantifiable numeric inputs from fundus images via sophisticated image pre-processing and segmentation pipelines. Creating an effective and automated biomarker extraction workflow for validation with established clinical observations.
- Development of a hybrid neural network architecture that combines the advantages of both ViTs and CNNs. By leveraging intra-branch self-attention and an inter-branch bidirectional cross-attention based fusion technique, the proposed model facilitates feature alignment and contextual refinement to aid with multimodal glaucoma classification.
- Benchmarking and comparing the performance of the model, thereby validating its effectiveness and exploring how varying data modalities influence outcomes.
- Providing empirical evidence demonstrating the complementary nature of the extracted clinical biomarkers through ablation studies [4.2], statistical analysis [4.3] and model interpretability [4.5].

2. Related work

Recent advances in deep learning have led to widespread adoption employing such techniques for the classification of glaucoma from retinal fundus images. The use of CNNs and ViTs for image classification, along with models for the optic disc, optic cup, and retinal vasculature segmentation, have gained rapid popularity. Shyamalee and Meedeniya [11] investigate the use of three different CNN architectures for direct glaucoma classification from retinal fundus images, demonstrating the effectiveness of deep learning models in identifying glaucoma-related patterns. Sudhan et al. [12] perform optic cup segmentation using U-Net. Transfer learning is then leveraged using the DenseNet-201 [13] model for feature extraction and a Deep Convolutional Neural Network (DCNN) to perform the final classification. The model achieves a testing accuracy of 96.90% on the Online Retinal Fundus Image Database for Glaucoma Analysis and Research (ORIGA)-light [14] dataset. A limitation of this approach is that it does not account for generalizability of the model across different datasets, but instead restricts testing to only a single dataset.

Haja et al. [15], Magboo et al. [16], and Diaz-Pinto [17] use CNNs for glaucoma classification, highlighting their ability to identify changes in the optic nerve head and optic cup from fundus images. While CNNs perform well on image classification tasks, they are limited by the size of their local receptive fields and struggle to accurately capture long-range dependencies. The limited ability of CNN-based models to capture global contextual relationships, combined with the potential loss of precise spatial information, can reduce their effectiveness in accurately identifying glaucoma-related patterns. As a result, these limitations may negatively affect diagnostic accuracy in glaucoma classification. Addressing these limitations, Sivakumar and Penkova [18] employ Swin transformers which outperform existing CNN models and prove to be better for glaucoma classification.

Vision transformers make use of a self-attention mechanism which allows them to capture global dependencies in the image. Compared to CNNs, ViTs are able to scale better as the size of the dataset increases. Due to their architecture and ability to capture broad contexts from the image, Sivakumar and Penkova [18] and Singh et al. [19] use vision transformers for glaucoma classification. The performance of the model is benchmarked against various optimizers such as Adamax, SGD, RMSProp, Adagrad, etc. Sakunthala et al. [20] compare the performance of ViT and Swin transformers with traditional CNN models.

Changes in the optic cup are one of the primary indications of glaucoma. Prediction of glaucoma based on variations in the optic cup structure is a task of vital importance. Shyamalee and Meedeniya [21] propose a segmentation-based approach using Attention U-Net models with three different CNN backbones, namely Inception-v3, VGG19, and ResNet50, for optic disc and optic cup segmentation from retinal fundus images. Zhou et al. [22] propose a novel approach involving a Locally Statistical Active Contour Model using a Structure Prior (LSACM-SP) to segment both the optic disc and the optic cup. Their method achieves an accuracy of 89.01% and 84.69% on the DRISHTI-GS [23] and RIM-ONE r2 [24] datasets, respectively. Chen et al. [25] proposed the use of the YOLOv8 model for optic disc and cup segmentation on the REFUGE dataset [26]. Shyamalee and Meedeniya [27] propose a hybrid methodology that performs optic disc and optic cup segmentation followed by glaucoma classification. In their approach, the Attention U-Net model with a ResNet50 encoder achieves an optic disc segmentation accuracy of 99.58%, demonstrating the effectiveness of combining segmentation and classification for glaucoma detection. Kim et al. [28] make use of FCNs (Fully Convolutional Networks) along with U-Net to segment the optic disc and cup from a particular ROI.

Relying solely on image-based data may not be sufficient to adequately classify glaucoma. Numeric glaucoma biomarkers are essential to validate existing hypotheses regarding whether a patient has glaucoma. Tsai et al. [29] emphasize that depending on a single clinical biomarker for glaucoma classification is insufficient. Instead, employing a combination of multiple biomarkers and tests is more effective for the rapid diagnosis of glaucoma.

Each of the methodologies explored provides certain advantages for glaucoma classification, however, they fail to overcome inherent limitations of not accounting for data from different modalities. Moreover, different features in the form of glaucoma biomarkers and their integration with features from other modalities have not been thoroughly explored in the context of glaucoma classification. Additionally, the absence of reliable, quantifiable feature-extraction pipelines can make model predictions less transparent and harder to interpret. This lack of interpretability may reduce clinicians' confidence in automated diagnostic systems, thereby limiting their practical clinical utility and widespread adoption. The proposed study focuses on segmentation based feature extraction of glaucoma biomarkers and overcomes the

limitations of existing techniques by using 3 different transformer models in conjunction with the extracted biomarkers. Features in the form of glaucoma biomarkers such as retinal diameter, optic disc diameter, optic cup diameter, macular diameter, optic cup to disc ratio, and retinal vasculature density provide valuable information for the accurate diagnosis of glaucoma. The proposed methodology evaluates the efficacy of combining both images and glaucoma biomarkers for multimodal glaucoma classification.

3. Methodology

3.1. Dataset used

The multimodal model is trained on the Standardized Multi-Channel Dataset for Glaucoma (SMDG-19) [30], a comprehensive resource that comprises 19 publicly available glaucoma datasets [31], as detailed in Table 1. The training, validation and testing of the multimodal model is performed on stratified splits in an 80:10:10 ratio, where the samples are drawn from the entirety of the SMDG-19 corpus. The evaluation protocols followed throughout this paper are consistent with those observed in contemporary glaucoma classification literature such as de Souza et al. [32], and Chincholi et al. [33], where the models were trained and evaluated on the same dataset. This study incorporates both non-glaucoma images (class 0) and glaucoma images (class 1) to effectively train the hybrid multimodal model. The dataset features rich fundus images paired with precise segmentation masks for the optic cup, optic disc, blood vessels, arteries, and veins. Moreover, Spectral Optical Coherence Tomography (OCT) images along with their corresponding optic cup and optic disc segmentation masks are included as part of this dataset. In addition to the images, the dataset provides limited metadata that reveals demographic information about patients. The primary objective of this dataset is to drive exploration and open research in the field of glaucoma. The segmentation based biomarker extraction was used to evaluate whether there was a significant difference in the optic disc and cup diameters of patients with and without glaucoma. All images used for training segmentation models were identified and removed from the SMDG-19 dataset prior to train:validation:test splitting to eliminate sample-level leakage. The transformer backbones used in the multimodal classification model are pretrained on ImageNet, which contains no fundus or medical images, thereby preventing representation-level leakage through the image modality. Additionally, the extracted biomarkers are explicitly defined as deterministic geometric measurements (e.g., ratios, areas, circumferences) derived from segmentation masks, rather than learned feature embeddings. These measurements do not encode dataset-specific distributional information and therefore cannot introduce representation leakage. The segmentation pipeline thus serves only to perform structural quantification, ensuring a clear separation between preprocessing and classification. To ensure clarity and relevance, the images and segmentation masks in the dataset have been standardized and processed to optimize the retention of the image foreground. Background cropping for SMDG-19 [30] has been performed to discard irrelevant information along with standardization involving image centering, addressing instances of missing data with padding and resizing them to a uniform 512 × 512 pixels.

Table 1
Datasets used to create SMDG-19 (Sivakumar and Penkova [18]).

Dataset	0 (No. Glaucoma)	1 (Glaucoma)
BEH [34]	463	171
CRFO-v4 [35–37]	31	48
DR-HAGIS [38]	0	10
DRISHTI-GS1-TRAIN [23]	18	32
DRISHTI-GS1-TEST [23]	13	38
EyePACS-AIROGS [39]	0	3269
FIVES [40]	200	200
G1020 [41]	724	296
HRF [42]	15	15
JSIEC-1000 [43]	38	0
LES-AV [44,45]	11	11
OIA-ODIR-TRAIN [46,47]	2932	197
OIA-ODIR-TEST-ONLINE [46,47]	802	58
OIA-ODIR-TEST-OFFLINE [46,47]	417	36
ORIGA-light [14]	482	168
PAPILA [48]	333	87
REFUGE1-TRAIN [26]	360	40
REFUGE1-VALIDATION [26]	360	40
sjchoi86-HRF [49]	300	101
Total	7499	4817

3.2. Optic disc diameter calculation

A sample of 2805 images containing optic disc segmentation masks was selected from the SMDG dataset. These samples were divided to optimize the model training, 80% were designated for the training set, 10% for the validation set, and the remaining 10% for testing. This division yielded 2244 images in the training set, 280 in the validation set, and 281 in the test set. Each image was carefully resized to 128 × 128 pixels and normalized for consistency. To significantly enhance the generalization capability of the deep learning model, augmentation techniques were employed. These included horizontal flipping, vertical flipping, random rotation by 90°, and shift-scale-rotation applied to both the original retinal fundus images and their corresponding optic disc segmentation masks. The applied augmentation techniques help prevent the model from memorizing the training data and encourage it to learn structural features present in retinal fundus images. These techniques are particularly relevant for glaucoma classification because they allow the model to learn retinal structures regardless of their exact position or orientation within the image. For example, shifting enables the model to detect the optic disc even when it is not perfectly centered in the image, while scaling helps account for variations in magnification caused by different imaging devices or eye sizes. Rotation augmentation further improves resilience to slight misalignments that may occur during image acquisition. The performance of the U-Net model was evaluated using key metrics such as accuracy, Dice coefficient scores, and IoU (Intersection over Union) values.

The integration of Binary Cross Entropy (BCE) loss with Dice loss significantly enhances the versatility of the model, capitalizing on the ability of Dice loss to handle overlaps. Binary Cross Entropy (BCE) loss operates at the pixel level, providing stable gradients and encouraging accurate probabilistic predictions for each pixel. However, BCE alone may not effectively capture the spatial overlap between predicted and ground truth regions, especially in cases of class imbalance. Dice loss addresses this limitation by directly optimizing the overlap between predicted and target segmentation regions. By combining these two loss functions (BCE-Dice loss), the model benefits from both stable pixel-wise learning and improved region-level segmentation accuracy, resulting in enhanced performance. The model was trained for 30 epochs using the combined BCE-Dice loss function. We employed an Adam optimizer with a learning rate of 1×10^{-3} paired with a scheduler designed to reduce the learning rate on plateaus, whenever the Dice score does not increase for 3 epochs. The results from training the U-Net model for optic disc segmentation are compellingly showcased in Table 2 and Fig. 1.

Once the model training is completed, we apply contour detection to the resulting predicted mask. The diameter of the largest minimum enclosing circle traced around that contour accurately reveals the diameter of the optic disc, offering a precise measurement essential for further analysis.

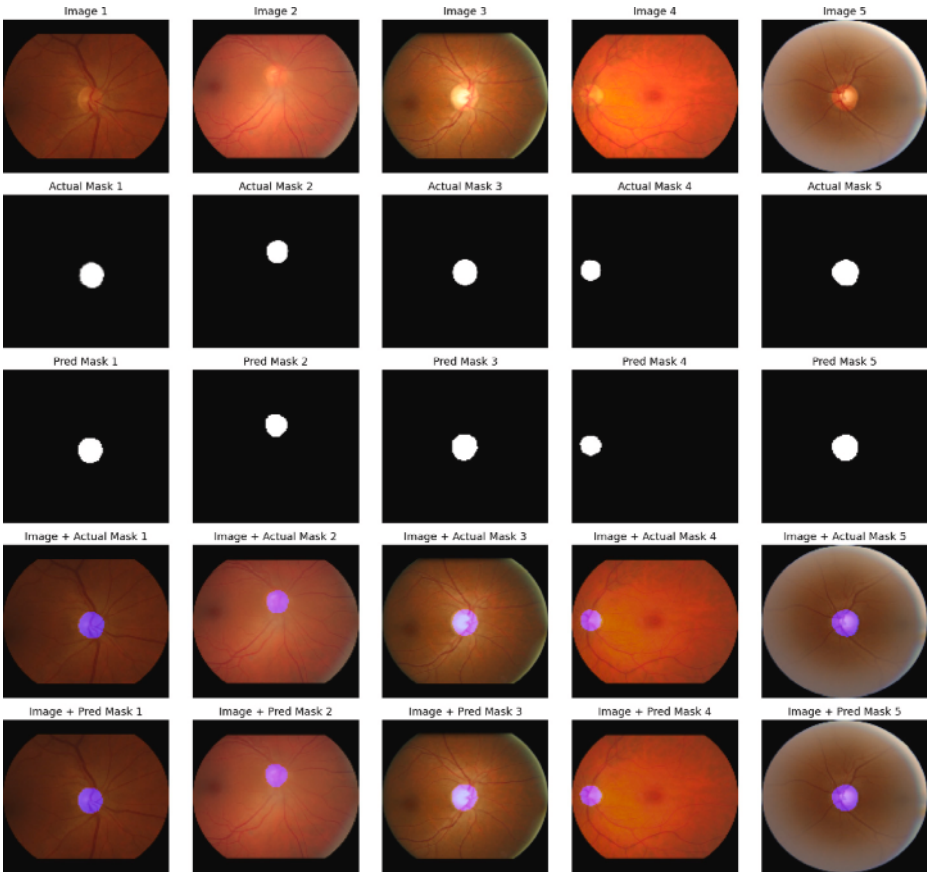


Fig. 1. Actual and predicted optic disc segmentation masks.

Table 2
Evaluation metrics for optic disc segmentation with U-Net.

Set	Loss	Dice coefficient	IoU	Accuracy
Train	0.07527	0.93702	0.88167	0.99641
Validation	0.07005	0.94137	0.88934	0.99658
Test	0.08081	0.93305	0.87495	0.99612

3.3. Optic cup diameter calculation

A comprehensive selection of 2805 fundus images was made from the SMDG dataset, strategically dividing the data into 80% for the training set, 10% for validation, and 10% for testing. Every aspect of our methodology, including data preprocessing steps, data augmentation methods, model architecture, evaluation metrics, the loss function, the optimizer and the scheduler aligns with those used in the optic disc segmentation [3.2]. The U-Net model designed for the optic cup segmentation underwent rigorous training over 50 epochs with a learning rate of 1×10^{-3}. The results, depicted clearly in Table 3, as well as Fig. 2 underscore the effectiveness of our approach.

The rationale behind training the optic cup segmentation model for 50 epochs lies in the fact that the optic cup is a smaller region inside the optic disc with low contrast boundaries and a greater anatomical variability. Due to this, the optic cup segmentation model was trained for 50 epochs to allow the model to accurately learn and refine boundary representations, whereas the optic disc segmentation model [3.2] was able to reach convergence in just 30 epochs.

The trained U-Net model is employed to generate segmentation masks for the optic cup in the retinal fundus images. Subsequently, contour detection is performed on the predicted mask, and a minimum enclosing circle is precisely fitted over the largest contour. The diameter of this circle provides the diameter of the segmented optic cup.

While prior approaches often rely solely on images or isolated handcrafted features, our methodology explicitly integrates quantitative anatomical measurements such as optic disc and cup diameters and cup-to-disc ratios, with deep learning-based feature representations. The ablation study [4.2] shows that these learned clinical biomarkers outperform standalone vision transformer features in terms of test accuracy and ROC-AUC. Neither deep learning features nor numerical biomarkers alone achieve optimal performance, however, the proposed fusion of both modalities delivers superior results on the test set. The performance gains from incorporating numerically precise and clinically meaningful biomarkers are statistically significant ($p = 4.13124 \times 10^{-31}$) according to DeLong's test (Table 14). These results underscore the critical importance of extracting and integrating anatomical biomarkers, demonstrating that their inclusion provides meaningful improvements beyond incremental image-based approaches.

3.4. Retinal vasculature density calculation

A total of 462 images were carefully selected from the SMDG dataset with the training split accounting for 80% of the total dataset size, containing a total of 369 images. The remaining 10% of the images were allocated to the validation set and the final 10% of the images were used as part of the test set. To improve the performance of the model and reduce overfitting, data augmentation techniques such

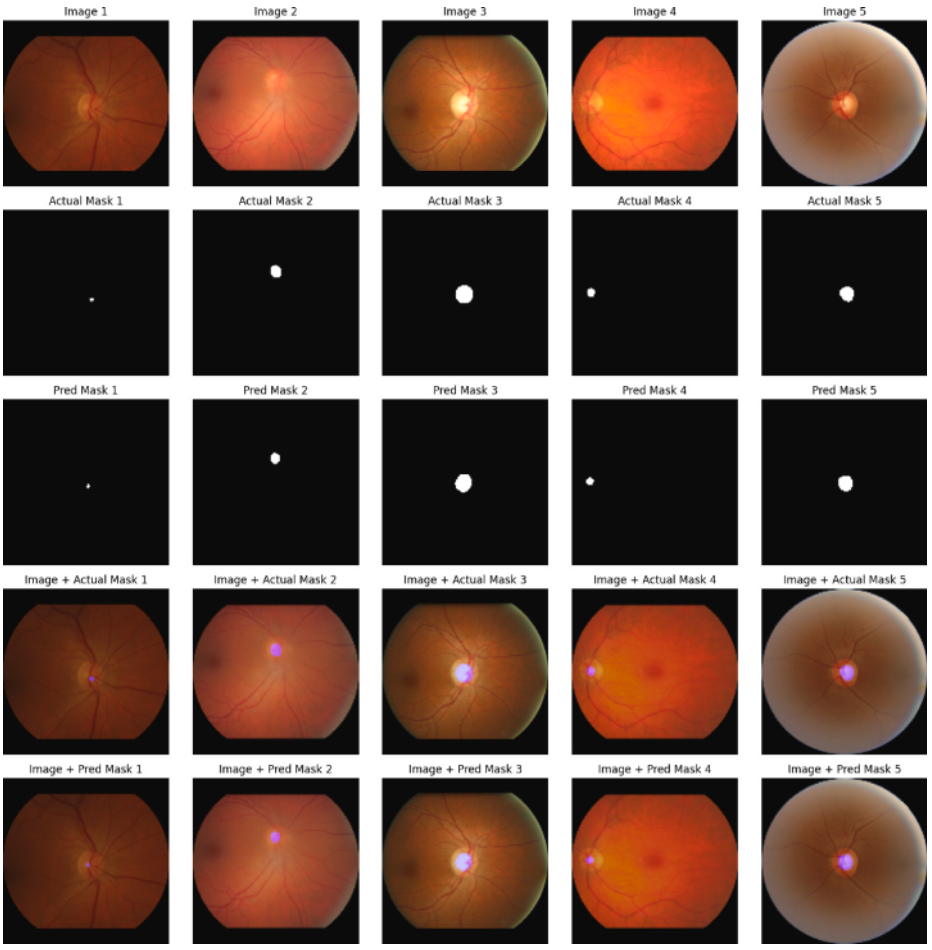


Fig. 2. Actual and predicted optic cup segmentation masks.

Table 3
Evaluation metrics for optic cup segmentation with U-Net.

Set	Loss	Dice coefficient	IoU	Accuracy
Train	0.17678	0.83486	0.71769	0.99770
Validation	0.19061	0.82147	0.69828	0.99775
Test	0.18292	0.82877	0.70913	0.99758

as horizontal image flipping, vertical image flipping, random rotation by 90°, shift scale rotation by 15°, random contrasting brightness, Gaussian blurring, Gaussian noise and hue saturation were applied to the images of the training set. These data augmentation techniques effectively increased the total number of training samples to 1107 images. Flipping and rotation augmentations help the model consistently identify vascular structures even when images appear in different orientations. Gaussian blurring and noise simulate variations in image quality and acquisition conditions, enabling the model to learn more vascular patterns. Additionally, hue and saturation adjustments account for color variability across different imaging devices and lighting conditions, reducing the model's reliance on specific color distributions. All the images in the dataset were normalized and resized to 128 × 128 pixels. The loss function, scheduler, optimizer and evaluation metrics excatly match those used for the optic disc segmentation [3.2]. The model was trained for 50 epochs using a learning rate of 1×10^{-3}. The results of the retinal vasculature segmentation are depicted in Table 4 and Fig. 3.

The retinal vasculature density was computed by calculating the total number of pixels occupied by the vasculature segmentation mask and dividing it by the total number of pixels present in the retinal fundus image.

3.5. Macular area calculation

A subset of 728 images was randomly selected from the SMDG dataset. These images were manually annotated using online tools available on Robof-low. The number of images in the dataset was tripled to 2184 by applying auto-orientation of pixel data (with EXIF-orientation stripping), horizontal image flipping with a 50% probability and salt and pepper noise to 1% of the pixels in the image. Image flipping helps the model better learn the spatial relationships between retinal regions and the macular area, improving its ability to detect relevant anatomical structures. The addition of salt-and-pepper noise simulates low-quality imaging conditions and enhances the model's ability to handle noise and other artifacts commonly present in real-world fundus images. The dataset was partitioned into a training set containing 80% of the images, a validation set comprising 10% of the images and a testing set containing the remaining 10% of the images. All the images in the dataset were resized to 128 × 128 pixels and normalized for training. The SegFormer model [50] was used as the network backbone and trained for 20 epochs with a learning rate of 1×10^{-4} using an Adam optimizer and a cosine annealing scheduler. The architecture of the custom SegFormer model used for the segmentation

Table 4
Evaluation metrics for retinal vasculature segmentation with U-Net.

Set	Loss	Dice coefficient	IoU	Accuracy
Train	0.32766	0.80119	0.66856	0.93257
Validation	0.32374	0.79800	0.66417	0.93360
Test	0.33550	0.78687	0.64882	0.93373

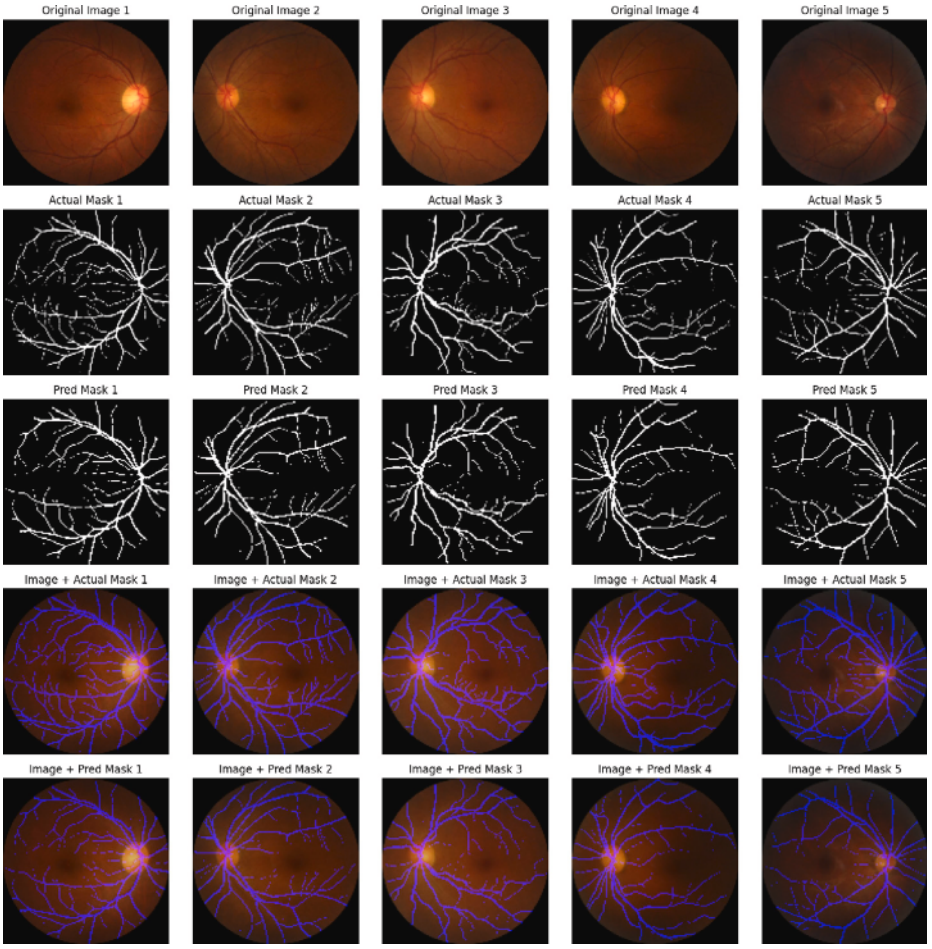


Fig. 3. Actual and predicted retinal vasculature segmentation masks.

task is depicted in Fig. 4. The output from the SegFormer backbone are passed to the Feature Pyramid Network (FPN) and Squeeze-and-Excitation (SE) blocks of the neural network. The FPN component of the network allows for the generation of semantically rich feature maps at a high level. The SE block dynamically assigns importance to each channel and adaptively recalibrates channel-wise feature responses. This is highly effective in overcoming the limitations of traditional CNNs where each channel is assigned an equal amount of importance for various downstream tasks. The inputs from the FPN and SE block are passed to the fusion and residual layers consisting of 2 convolutional, batch normalization and ReLU layers separated by a dropout layer with a dropout rate of 0.3. A residual connection consisting of a convolutional skip connection is present to ensure that the input features can contribute directly to the output. The fusion of rich feature representations from these branches is then directly passed to the decoder head consisting of 2 convolutional layers with a batch normalization, ReLU and dropout layer in between. A dropout rate of 0.4 is used as part of the decoder head to enhance regularization strength and prevent overfitting. The output from the decoder head is subsequently passed to the refinement module containing a convolutional layer, a batch normalization layer, a ReLU activation layer, a dropout layer with a dropout rate of 0.2 and lastly a convolutional layer. The refinement block enables the model to learn a semantically richer segmentation map by refining the output of the decoder head using 2 convolutional layers. The final outputs from the refinement module are used to predict the segmentation masks for the macula.

The results of the macular area segmentation are shown in Table 5, and Fig. 5. The macular area is computed by calculating the sum of all the pixels occupied by the predicted segmentation mask. The macular area can then be used to compute the radius, diameter and circumference of the macular region.

3.6. Tabular classification with machine learning

The retinal diameter (d) is extracted from the fundus images in our dataset to evaluate whether significant differences exist in the retinal diameters between patients with and those without glaucoma. Rim et al. [51] use spectral domain OCTs to gauge the average luminal diameters of retinal arteries and veins, calculating them to be 95.1 ± 16.1 μm and 132.6 ± 17.8 μm, respectively. Yoo et al. [52] make use of retinal vessel diameters to distinguish between eyes with and without Open Angle Glaucoma (OAG). Other existing approaches [53–55] focus

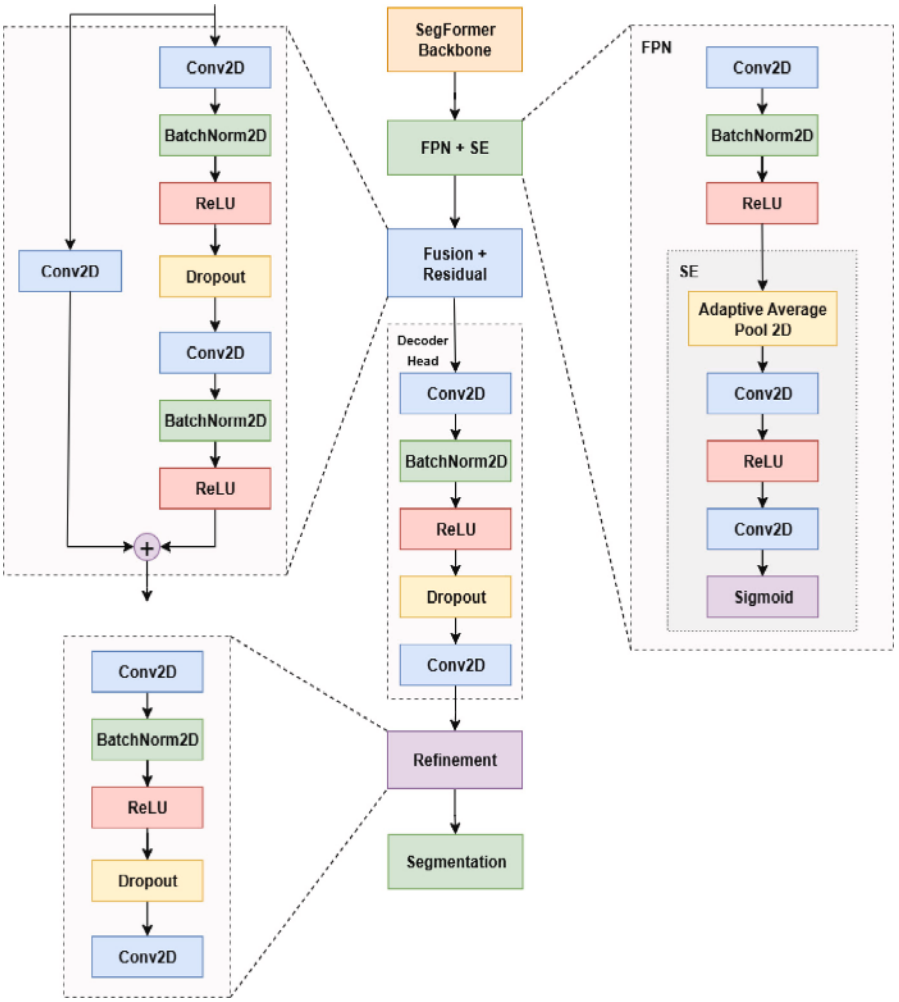


Fig. 4. Custom SegFormer architecture for macular segmentation.

Table 5
Evaluation metrics for macular area segmentation with SegFormer.

Set	Loss	Dice coefficient	IoU	Accuracy
Train	0.08606	0.91400	0.84551	0.99503
Validation	0.09317	0.91364	0.84571	0.99441
Test	0.08874	0.91580	0.84877	0.99499

on calculating retinal vessel diameters, however, do not take into account the effect of retinal diameter size on the prediction of glaucoma from fundus images. The proposed study seeks to addresses these limitations by extracting the retinal diameter and employing feature engineering techniques. This involves calculating the retinal radius, retinal area, retinal circumference and creating ratios between features extracted from segmentation of the optic disc [3.2] and optic cup [3.3] of the eyes. Edges in the fundus images are extracted using Canny edge detection [56]. Canny edge detection is sensitive to parameter settings and may fail to capture weak or poorly defined edges, such as the retinal boundary in some fundus images. As a result, important structural details may occasionally be missed. To mitigate this limitation, Canny edge detection is followed by contour detection, which helps identify continuous boundaries and improves the detection of partially defined retinal edges. After extracting the contours, a minimum enclosing circle is fitted around the largest detected contour to approximate the retinal boundary and estimate the retinal diameter. However, the accuracy of this method can be affected when the retinal boundary is not clearly defined or when noise and artifacts are present in the image. By incorporating contour detection as an intermediate step between Canny edge detection and the minimum enclosing circle algorithm, the retinal boundary becomes more clearly defined, enabling the minimum enclosing circle method to more reliably estimate the retinal circumference and diameter. By utilizing the center of this minimum enclosing circle along with the identified retinal boundary, the diameter of the retina is calculated in terms of pixels (px), as illustrated in Fig. 6.

To ensure the utmost integrity of our data and prevent any possibility of leakage, we removed all samples used for training and evaluating the U-Net models focused on optic disc and optic cup segmentation. Furthermore, we have excluded all samples containing blood vessel segmentation masks. As a result, the total number of samples has been reduced from 12316 to 8818 in the final dataset, enhancing its quality and reliability. Once we extract the diameters of the retina, optic disc and optic cup from the images, we create new features that are subsequently passed to advanced machine learning models for tabular classification as shown in Fig. 7.

Based on the retinal, optic disc and optic cup diameters extracted from the images, we have constructed the features employed by the

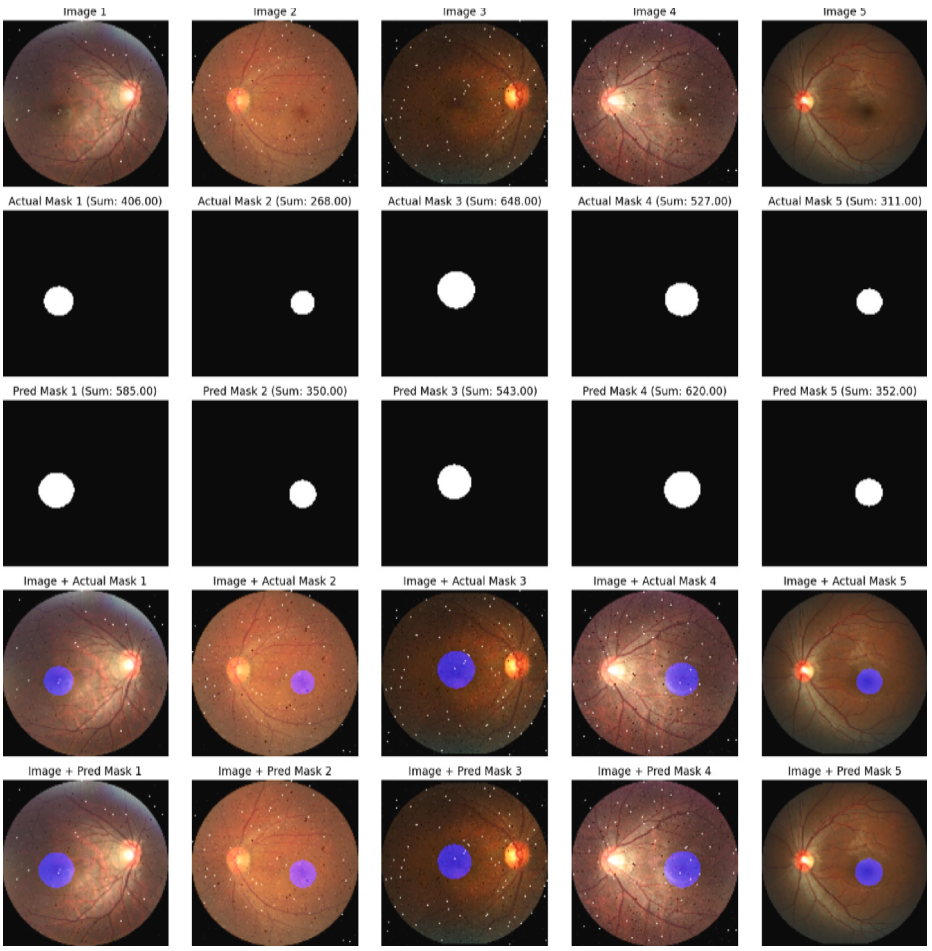


Fig. 5. Actual and predicted macular segmentation masks.

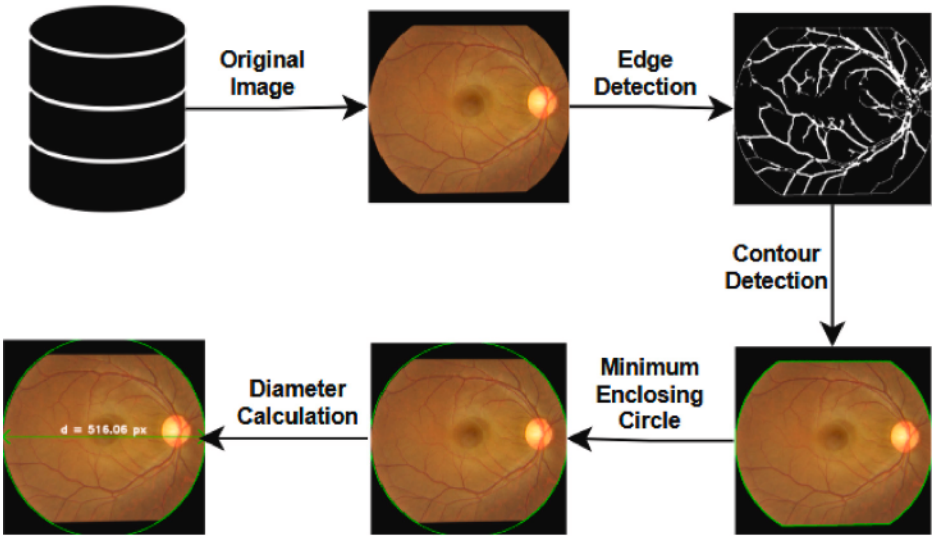


Fig. 6. Steps depicting the calculation of the retinal diameter.

machine learning models, as detailed in Table 6. To ensure data integrity, all missing values in the dataset have been effectively addressed by replacing them with the mean of their respective columns. We implemented a strategic train-validation-test split of 80:10:10 on the dataset, resulting in a train set of 7054 samples, a validation set of 882 samples, and a test set that comprises the remaining 882 samples. The impressive results obtained after training the machine learning models are showcased in Table 8. The chosen models cover a diverse spectrum

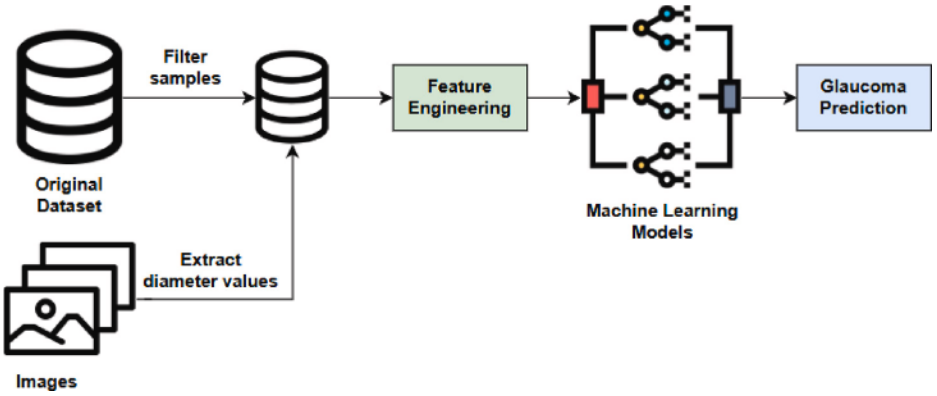


Fig. 7. Feature extraction and tabular classification pipeline.

of machine learning approaches, including linear classifiers, tree-based models, ensemble methods, and state-of-the-art gradient boosting techniques. Logistic regression is capable of modeling linear relationships and probabilities, whereas decision trees work well with data that is not linearly separable. Random forest was selected for its versatility, XGBoost was selected for its ability to prevent overfitting, whereas histogram gradient boosting was chosen for its rapid training speed. Lastly, CatBoost was chosen for its permutation-based approach to handle prediction shift. The goal of training machine learning models is to show the accuracy limitations that arise from relying solely on a numeric tabular driven approach for glaucoma classification. Overall, this selection demonstrates the strengths and limitations of tabular machine learning models for glaucoma classification, highlighting the challenges of utilizing only numeric data without complementary image-based information.

To assess the impact of glaucoma, we present a compelling comparison between patients with and without glaucoma in Table 7, highlighting the mean and standard deviation of key measurements. The average retinal diameter [57] typically ranges from 30 millimeters (mm) to 40 mm. For the purposes of our analysis, we assume a retinal diameter of 35 mm represents the mean retinal diameter of a non-glaucomatous patient corresponding to 513.38307 px in the fundus images. This allows us to establish that 1 px on the fundus image corresponds to 0.06818 mm in a healthy retina. To ensure the fairness of our statistical analysis, we meticulously exclude any values that fall below or exceed 1.5 times the Interquartile Range (IQR).

Crowston et al. [58] analyzed 3654 individuals and reported a median optic disc diameter of 1.50 mm, with 98.9% of discs ranging from 1.2 to 1.9 mm. The median cup-to-disc ratio was approximately 0.43, with the 95th percentile reaching 0.67 in patients with glaucoma. The statistics extracted from our dataset for patients with and without glaucoma (Table 7) closely match these clinically observed values.

It is important to note that the differences in retinal and optic disc diameters between patients with and without glaucoma is negligible. However, patients with glaucoma tend to exhibit slightly larger optic cup diameters than those without the condition. This increase in the diameter is a strong indication of glaucoma and leads to a corresponding increase in the optic cup-to-disc-ratio among patients with glaucoma. These results are consistent with the generally observed trend of glaucomatous optic cups being enlarged [45]. The enlargement in the cup-to-disc-ratio, also termed cupping, arises from elevated intraocular pressure (IOP) or compromised blood flow to the optic nerve. As the nerve fibers gradually degenerate, cupping and its associated effects tend to become more prominent in patients suffering from glaucoma. Similarly, the mean macular diameter extracted from patients without glaucoma (≈1.5 mm) aligns with reported clinical measurements [59], with minor differences attributable to manual annotation and the lack of publicly available standardized datasets. By benchmarking these extracted anatomical features against clinically verified data, we demonstrate the reliability of the multimodal features used in our model. Furthermore, the proposed multimodal model has been deployed [4.4] in a manner that allows for validation and potential adoption by medical experts, facilitating its use in real-world clinical settings.

Table 6
Features utilized by the machine learning models.

Feature name	Type (extracted or engineered)
Retina diameter	Extracted
Optic disc diameter	Extracted
Optic cup diameter	Extracted
Vasculature density	Extracted
Macular area	Extracted
Retina radius	Engineered
Retina area	Engineered
Retina circumference	Engineered
Optic cup radius	Engineered
Optic cup area	Engineered
Optic cup circumference	Engineered
Optic disc radius	Engineered
Optic disc area	Engineered
Macular radius	Engineered
Macular diameter	Engineered
Macular circumference	Engineered
Optic disc circumference	Engineered
Optic disc to retina diameter ratio	Engineered
Optic disc to retina area ratio	Engineered
Optic cup to disc diameter ratio	Engineered
Optic cup to disc area ratio	Engineered
Optic cup to retina diameter ratio	Engineered
Optic cup to retina area ratio	Engineered

3.7. Multimodal architecture

The architecture of the multimodal model is depicted in Fig. 8. Model 1 and Model 2 correspond to the Swin-b and Swin-V2-b models, respectively.

Swin-b was chosen for its hierarchical feature extraction and highly accurate feature pyramids, enabled by the shifted window attention mechanism, which improves local and global feature modeling. Its successor, Swin-V2-b, was included for its enhanced scalability, efficient

Table 7
Statistics for patients with and without glaucoma.

Measurement	Patients with glaucoma	Patients without glaucoma
Retinal Diameter (mm)	34.99090 ± 0.17833	35.00000 ± 0.18061
Optic Disc Diameter (mm)	1.38473 ± 0.14760	1.37436 ± 0.14622
Optic Cup Diameter (mm)	0.83414 ± 0.17096	0.68458 ± 0.15744
Optic Cup to Disc Ratio	0.60238 ± 1.15827	0.49811 ± 1.07673
Macular Diameter (mm)	1.74190 ± 0.27409	1.69769 ± 0.24326

Table 8
Performance of the machine learning models for tabular classification.

Model name	Test accuracy	Test ROC AUC	Test F1-score
Logistic Regression	0.74263	0.80559	0.74300
Decision Tree Classifier	0.66213	0.65747	0.66241
Random Forest Classifier	0.76077	0.83210	0.75969
XGBoost Classifier	0.75283	0.84025	0.75276
Histogram Gradient Boosting Classifier	0.76531	0.84023	0.76610
CatBoost Classifier	**0.77098**	**0.85580**	**0.77061**

computation, and improved learning stability through residual post-normalization, making it suitable for large-scale and high-resolution image analysis. Collectively, these transformers were chosen to leverage complementary strengths in capturing both global and local image features, enhancing the performance of the multimodal glaucoma classification framework. The advantage of using transformer-based models, such as Swin-b and Swin-V2-b lies in their ability to effectively capture long range dependencies among image patches and to thoroughly understand global contexts. The process begins by partitioning the image into patches, which are transformed into linear embeddings of the image before being processed by the encoder block of the model.

The Swin-b model divides images into patches and applies linear transformations to convert them into vectors. With a patch size of 4×4 and a design featuring 18 encoder blocks, Swin-b stands out by replacing the conventional MHA (Multi-Head Attention) block with a transformative attention mechanism. This mechanism alternates between Window-based Multi-Head Self-Attention (W-MSA) and Shifted Window Multi-Head Self-Attention (SW-MSA) in consecutive encoder blocks. W-MSA segments the image into non-overlapping windows, applying the self-attention mechanism within each discrete region, rather than across the entire image. On the other hand, SW-MSA shifts the window at every alternate encoder block, ensuring that the model does not restrict itself to learning only local contexts within fixed windows partitions. The Swin-b model effectively integrates hierarchical learning with cross-window information retention, thereby enhancing glaucoma detection from fundus images.

The Swin-V2-b model uses image patches of size 4×4 and consists of 18 encoder blocks as part of its architecture. The final layer normalization block used in the encoder alleviates the vanishing and exploding gradients problem that might potentially arise during training of the neural network.

The features extracted from the retinal fundus images are concatenated after being independently extracted from the 2 transformer models. The concatenation helps achieve better image-based feature representations, leveraging the strengths of both the models, resulting in a more comprehensive understanding of the data.

To improve the ability of the model to generalize, data augmentation methods such as random horizontal flipping, random rotation by 10° and color jitter were applied. Furthermore, the images were resized to 224×224 and normalized before being fed into the multimodal model. To enhance the performance of the model, numeric features from the tabular dataset are scaled before being passed as an input to the multimodal model.

The multimodal architecture is structured into four stages. First, visually rich semantic features are extracted using two Swin Transformers, alongside the extraction of key clinical biomarkers. Next, the extracted features are projected into a shared embedding space to enable alignment across modalities. In the third stage, bidirectional cross-modal attention is applied, allowing image representations to attend to numerical features and vice versa. Finally, the fused embeddings are passed through a classification head to perform glaucoma detection. The proposed architecture explicitly models both intra-modal and inter-modal dependencies, rather than combining outputs post hoc. The numeric branch is implemented as a 2 block 1D-CNN pipeline arranged sequentially, where each block consists of a 1D convolutional layer, batch normalization, ReLU activation and max pooling. The blocks do not share parameters and progressively learn higher level representations of the input biomarkers. The resulting feature maps are then flattened and passed through fully connected layers to produce a fixed length feature vector. This vector serves as the final representation of the numeric branch.

Before applying the attention mechanisms, the feature representations from both the image and numeric branches are linearly projected into a shared embedding space of equal dimensionality. This projection ensures that the two modalities are compatible for subsequent self attention and cross attention operations. By aligning the feature dimensions, the model can effectively compute attention scores between modalities, enabling meaningful interaction and fusion. Self attention is applied within each modality to refine unimodal representations, and cross attention is then applied bidirectionally (image to numeric and numeric to image) to align complementary features. This allows the model to focus on regions consistent with high risk indicators for glaucoma classification.

Bidirectional cross attention is used to enable mutual interaction between modalities rather than a one-sided dependency. When the image features query numeric values, the model uses clinical biomarkers to guide attention towards relevant visual representations. Conversely, when numeric features query image embeddings, the model refines tabular representations using contextual information from the fundus images. This design ensures that both modalities contribute equally and prevent one modality from dominating the learning process. Since glaucoma diagnosis depends on both structural image cues and quantitative biomarkers, this symmetric interaction allows the model to learn more comprehensive and complementary feature representations than a single self attention mechanism.

The outputs from the cross MHA blocks are passed to a FFN (Feedforward Network) consisting of 2 linear layers separated by a GELU (Gaussian Error Linear Unit) activation function and a dropout of 0.3. The main benefit of GELU is that it provides smoother gradients as compared to the ReLU activation function.

The fused multimodal representation is passed through a multi-layer feedforward classification head, which incorporates normalization and regularization to improve training stability. This classification head transforms the joint feature representation into the final prediction through a series of non-linear projections, culminating in a binary output for glaucoma detection.

The contribution of the proposed model lies at the interaction level rather than the component level. While individual modules are established, the proposed bidirectional cross-modal attention mechanism introduces a new functional paradigm in which each modality dynamically conditions the other modality's representation. Unlike conventional late-fusion approaches that assume conditional independence,

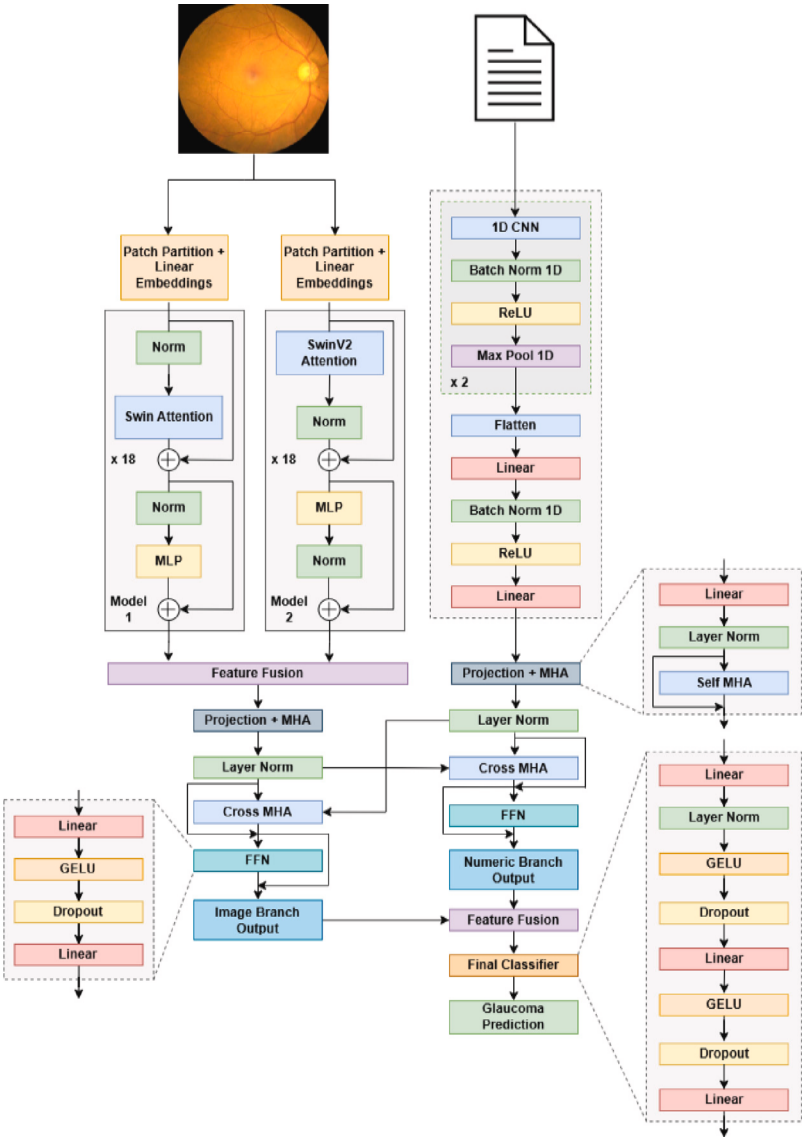


Fig. 8. Architecture of the multimodal model.

our method explicitly models cross-modal dependencies via symmetric attention, enabling iterative refinement within a shared embedding space. Ablation results [4.2] confirm that performance gains are attributable to this interaction mechanism rather than model scaling, while interpretability analysis [4.5] shows that clinically meaningful features emerge naturally from the learned attention. This suggests that the proposed design not only improves classification performance but also facilitates the learning of clinically consistent multimodal representations.

4. Results

4.1. Multimodal glaucoma classification model

The trained multimodal model was rigorously tested using performance metrics such as accuracy, precision, recall, F1-score, and ROC AUC. We leveraged an NVIDIA Tesla P100 GPU with a memory capacity of 16 GB, provided by Kaggle to train the model. The training took approximately 3 h on a single GPU. A cross entropy loss function was used during training along with an Adam optimizer having a learning rate of 1×10^{-4} and a weight decay of 1×10^{-5}. Weight decay prevents the model from overfitting by imposing a penalty on large weights, which ultimately promotes greater generalizability. To enhance the training process, a StepLR scheduler was used since it assists in decreasing the learning rate by a controlled amount with a step size of 2 and a gamma value of 0.1. Fig. 9 represents the model's performance on the test set, where it achieved a test accuracy of 92.63039%. The model achieves a precision of 0.90404, a recall of 0.92987 and an F1-score of 0.91677. Analysis of the confusion matrix in Fig. 9 reveals:

- True Negatives (459): These samples were correctly classified as healthy, indicating the model's accuracy in identifying non-glaucomatous retinas.
- True Positives (358): These images were correctly identified as glaucomatous, demonstrating the model's ability to accurately detect glaucoma cases.
- False Positives (38, Type I errors): Healthy retinas were misclassified as glaucomatous, leading to unnecessary follow-up or testing.
- False Negatives (27, Type II errors): Actual glaucoma cases were misclassified as healthy, representing missed diagnoses.

In the context of glaucoma screening, minimizing false negatives is particularly critical, as failing to detect actual cases could delay

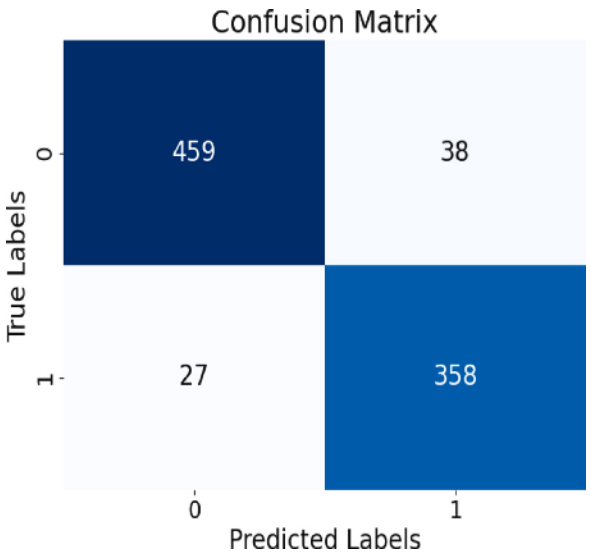


Fig. 9. Confusion matrix for the multimodal model.

treatment. Therefore, a high recall is prioritized to ensure that nearly all glaucoma cases are correctly identified from retinal fundus images, while maintaining a balance with precision to reduce unnecessary false alarms. Additionally, the model obtained an ROC AUC score of 0.97464 for both classes as demonstrated in Fig. 10. The ROC AUC score measures the model's ability to distinguish between glaucoma and non-glaucoma cases across all possible classification thresholds, reflecting the trade-off between the true positive rate and false positive rate. In practical terms, an ROC AUC of 0.97464 indicates that the multimodal model will assign a higher predicted probability to a randomly chosen glaucomatous retinal fundus image than to a randomly chosen healthy image approximately 97.5% of the time. This demonstrates excellent discriminative performance and indicates that the model is highly effective at distinguishing glaucomatous from non-glaucomatous eyes.

Table 9 benchmarks and compares the performance of the proposed multimodal model with existing approaches. Velpula et al. [60] leverage the power of 5 CNN models and use a fusion based approach to perform the final glaucoma classification. This approach results in an accuracy of only 85.180%. De Souza et al. [32] combine EfficientNet-B3 along with attention modules to enable the model to learn contextual cues from the images in the dataset. Their proposed methodology achieves an accuracy of 88.27% on the SMDG-19 dataset. Chincholi et al. [33] employ a DEtection TRansformer (DETR) model, however, do not account for multimodal data and detect glaucoma with an accuracy of 90.480%. Shyamalee et al. [61] make use of InceptionNetV3 along with ResNet50 based U-Net segmentation for glaucoma classification. The model achieves an accuracy of 98.97% on the RIM-ONE dataset, 97.24% on the ACRIMA dataset and 91.17% on the ORIGA dataset. Other approaches [12,62] train CNN models on the ORIGA dataset. Alasmari et al. [63], similarly makes use of this dataset which is a single homogeneous dataset created under controlled conditions. Moreover, this dataset contains only 650 images with a mere 168 images belonging to class 1 (glaucoma), indicating a non-trivial label imbalance. While these results are competitive, they are obtained by evaluating models under single dataset evaluation settings with relatively consistent imaging conditions. A more appropriate comparison can be seen with studies [32,33] that are evaluated on the same dataset.

In contrast, the proposed framework is trained on the SMDG-19 dataset, which aggregates 19 heterogeneous fundus datasets, introducing substantial variability in image quality and acquisition conditions. The proposed multimodal approach integrates segmentation-derived clinical glaucoma biomarkers with rich visual features extracted from images, combining numeric and visual modalities to capture both anatomical and structural information. This feature fusion improves model stability, and generalizability. The proposed model achieves a test accuracy of 92.63% (95% CI: 90.70%–94.33%) on SMDG-19 outperforming prior methods evaluated on the same benchmark and using the same evaluation protocols, including De Souza et al. [32] and Chincholi et al. [33].

We employ t-SNE to project the multimodal high dimensional fused feature representations into a 2 dimensional space for visualization of the clusters for classification of glaucoma. We use a perplexity value of 100 and run the t-SNE algorithm for 1000 iterations, achieving a KL divergence of 0.23726, indicating stable preservation of the local structure. As illustrated in Fig. 11 clearly separable clusters can be observed for samples with and without glaucoma. The formation of well defined clusters is suggestive of the fact that the proposed multimodal model accurately learns discriminative representations of the features. A partial overlap can be observed towards the center of the graph reflecting the similarities between samples and the inherent difficulty of the task.

4.2. Ablation study

To assess the contribution of each component, an ablation study was conducted by iteratively training and testing individual branches of the architecture shown in Fig. 8, while removing the remaining components. The contribution of each component is systematically evaluated by conducting an ablation study across three axes: individual unimodal models (ViT-b-16, Swin-b, Swin-V2-b, and 1D-CNN), combinations of models (Swin and ViT with 1D-CNN), and different fusion strategies. While not present in the multimodal architecture, we train a ViT-b-16 model as a baseline, to provide a reference in guiding architectural design decisions for the model. All models in the ablation study were trained from scratch for 20 epochs with a learning rate of 1×10^{-4} and a weight decay of 1×10^{-5}, from scratch, following the same training protocol as the full multimodal model [4.1].

Feature concatenation and individual data branches were removed as part of the first axis of the ablation study, and all the individual neural networks trained consisted of a final linear layer with 2 neurons representing the number of classes. The results, presented in Table 10, demonstrate the impact of each modality on glaucoma classification. The 1D-CNN using numeric clinical biomarkers achieves the highest accuracy, highlighting the importance of quantifiable glaucoma features. Transformer-based models capture visual patterns but cannot outperform the 1D-CNN alone due to the lack of numeric data. These findings emphasize the complementary nature of the modalities, where integrating both numeric biomarkers and image-based features in the proposed multimodal model leads to superior performance over unimodal approaches, validating the effectiveness of the weighted fusion strategy.

The second part of the ablation study involves evaluating the base models (ViT-b-16, Swin-b and Swin-V2-b) in conjunction with the 1D-CNN as shown in Table 11. Compared to the unimodal base models, a significant increase in accuracy is observed when the base models are used with the 1D-CNN. This highlights the importance of considering data from both the image and the numeric (clinical biomarker) modality. As observed, the combination of ViT-b-16 and the 1D-CNN results in the lowest performance. To improve the performance of the final model, we drop the lowest performing component (ViT-b-16) and keep both the Swin models in the final architecture.

To supplement our findings, we evaluate the performance of various fusion strategies and its overall impact on the final glaucoma classification in Table 12. The model was trained using both Swin architectures while varying the fusion strategies. As evident, the cross-modal attention based fusion technique achieves the highest test accuracy, test

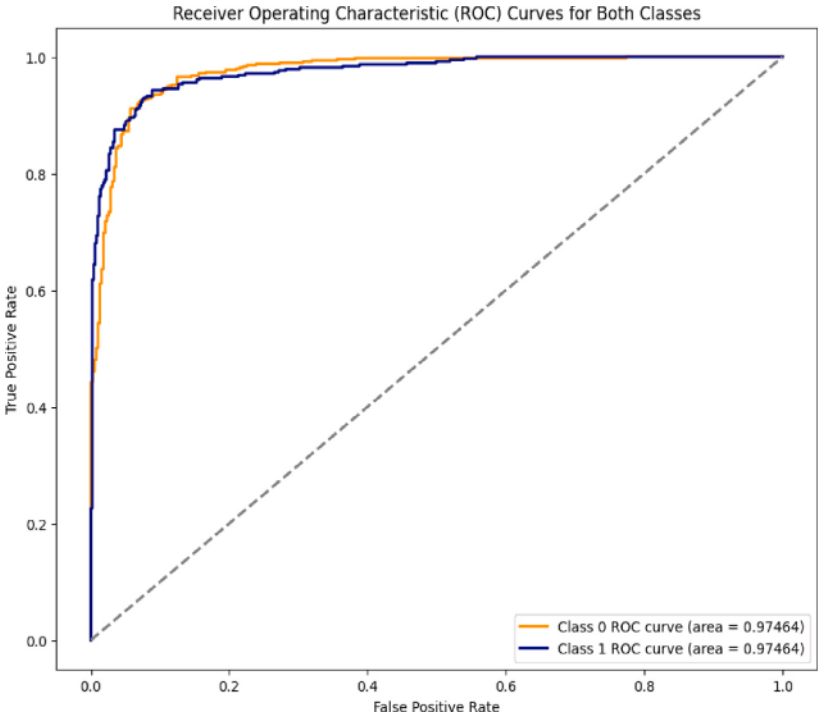


Fig. 10. Multimodal model ROC AUC curves.

Table 9
Performance comparison of various models.

Study	Dataset	Model	Accuracy
Alasmari et al. [63]	REFUGE [26] + ORIGA [14] + G1020 [41]	U-Net [7] + VGG16 [64]	84.00%
		MaskFormer [65] + Swin-b [9]	84.03%
Velpula et al. [60]	HVD [66] + DRISHTI [23]	Fusion (ResNet50 [6] + VGG19 [64] + AlexNet [67] + DenseNet-201 [13] + Inception-ResNet-V2 [68])	85.18%
Alasmari et al. [63]	REFUGE [26] + ORIGA [14] + G1020 [41]	U-Net [7] + ResNet50 [6]	86.00%
		U-Net [7] + MobileNet [69]	87.00%
De Souza et al. [32]	SMDG-19 [30]	EffAttNet (EfficientNet-B3 [70] + Attention Modules)	88.27%
Charry et al. [71]	DRISHTI [23] + RIM-ONE [24] (v1) + RIM-ONE [24] (v3)	DCNN + Morphometric Features + MLP	89.40%
Chincholi et al. [33]	SMDG-19 [30]	DETR (DEtection TRansformer) [72]	90.48%
Ours	**SMDG-19** [30]	**Swin-b** [9] + **Swin-V2-b** [10] + **1D-CNN**	**92.63%**

Table 10
Comparison of individual model performances in the ablation study (first axis).

Model	Test accuracy	Test precision	Test recall	Test F_1-Score	Test ROC AUC
ViT-b-16	0.74603	0.83128	0.52468	0.64331	0.80871
Swin-b	0.66327	0.73158	0.36104	0.48348	0.69893
Swin-V2-b	0.72222	0.76923	0.51948	0.62016	0.76082
1D-CNN	0.75510	0.72776	0.70130	0.71429	0.82188

precision, test F_1 score and test ROC AUC values. In contrast, alternative fusion approaches, including gated fusion, perform consistently worse.

Across the 3 axes of the ablation study, we observe that leveraging the 1D-CNN (trained on numeric biomarkers) and the Swin models (trained on fundus images) provide a significant increase in performance over traditional single modality base models. Moreover, using cross-modal attention improves the model's overall performance, demonstrating effective feature fusion and enhanced representation.

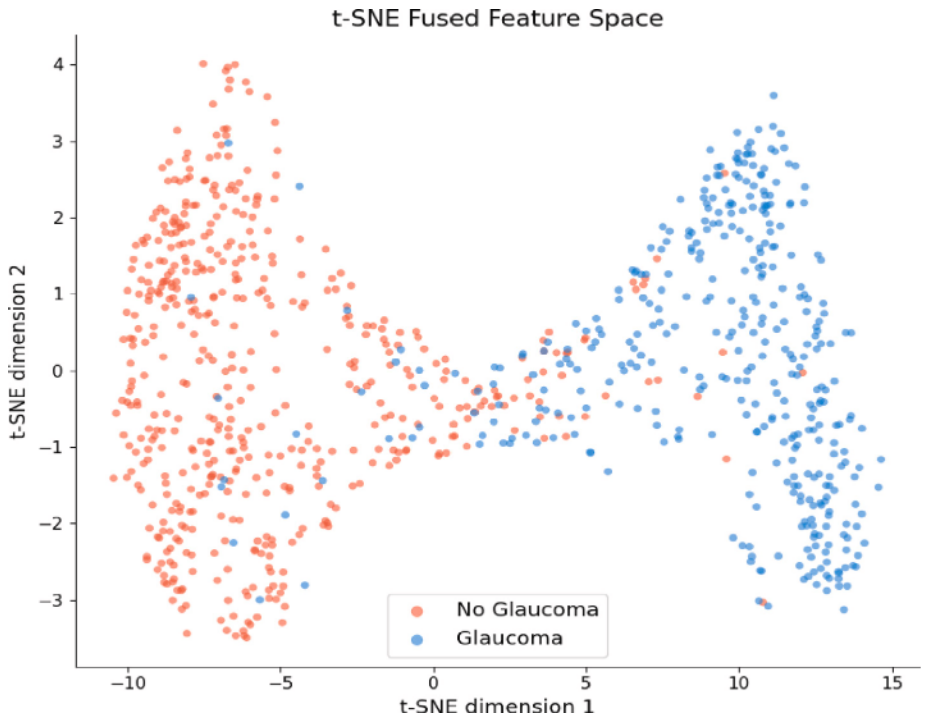


Fig. 11. t-SNE cluster separation.

Table 11
Comparison of base models + 1D-CNN in the ablation study (second axis).

Fusion strategy	Test accuracy	Test precision	Test recall	Test F_1-Score	Test ROC AUC
ViT-b-16 + 1D-CNN	0.90930	0.91553	0.87273	0.89362	0.96076
Swin-b + 1D-CNN	0.91500	0.87990	0.93247	0.90542	0.96844
Swin-V2-b + 1D-CNN	0.91517	0.91645	0.91169	0.91406	0.97275

Table 12
Comparison of various fusion strategies in the ablation study (third axis).

Fusion strategy	Test accuracy	Test precision	Test recall	Test F_1-Score	Test ROC AUC
Gated Fusion	0.80952	0.79245	0.76364	0.77778	0.86684
Feature Concatenation	0.91520	0.89776	0.93506	0.91603	0.97326
Cross-Modal Attention Fusion	0.92630	0.90404	0.92987	0.91677	0.97464

In addition to this, interpretability analysis [4.5] proves that both modalities focus on clinically meaningful features, including the optic cup, optic disc, vascular density, and retinal geometry. This indicates that the observed performance gains stem from effective cross-modal feature interactions rather than from simple feature aggregation.

4.3. Statistical analysis

Non-parametric bootstrapping with 1000 resamples of the test set was used to compute 95% confidence intervals (CI) for all evaluation metrics (Table 13). These intervals quantify the uncertainty of each metric and indicate the stability of model performance across resampled test sets. For the multimodal model, the CIs were narrow (e.g., Accuracy: 0.92630, 95% CI: 0.90703–0.94331; ROC AUC: 0.97464, 95% CI: 0.96467–0.98335), reflecting reliable and consistent predictions. In contrast, unimodal models like Swin-b exhibited wider CIs (e.g., test precision range of 0.12), highlighting higher variance in predictions. The multimodal integration mitigates this effect by combining complementary numeric and image-based features.

DeLong's test was applied to assess whether differences in ROC-AUC scores between models evaluated on the same test set are statistically significant, as shown in Table 14. A low p-value ($p < 0.05$) indicates a significant difference. For example, the multimodal model significantly outperforms the Swin-b model ($p = 1.74715 \times 10^{-54}$) confirming that integrating numeric biomarkers with image features leads to substantial performance gains. Among unimodal models, no significant difference was observed between ViT-b-16 and the 1D-CNN ($p = 4.7522 \times 10^{-1}$), indicating comparable ROC-AUC performance.

Table 15 shows the results of the McNemar test, used to measure the statistical significance between the performance of two classifiers and to check if the models being compared asymmetrically disagree in their predictions. Here, b represents cases where Model 1 is correct

Table 13
Lower and upper bounds for 95% confidence intervals across all model metrics.

Model	Test accuracy	Test precision	Test recall	Test F_1-Score	Test ROC AUC
Multimodal	[0.90703, 0.94331]	[0.87262, 0.93230]	[0.90261, 0.95480]	[0.89390, 0.93671]	[0.96467, 0.98335]
ViT-b-16	[0.71769, 0.77438]	[0.78365, 0.87805]	[0.47607, 0.57403]	[0.59999, 0.68703]	[0.78026, 0.83783]
Swin-b	[0.63379, 0.69388]	[0.67001, 0.79048]	[0.31382, 0.40788]	[0.43144, 0.53110]	[0.66389, 0.73417]
Swin-V2-b	[0.69615, 0.75060]	[0.71813, 0.81930]	[0.46858, 0.56488]	[0.57725, 0.66154]	[0.72827, 0.79311]
1D-CNN	[0.72562, 0.78345]	[0.68253, 0.76882]	[0.65262, 0.74746]	[0.67654, 0.74750]	[0.79366, 0.84788]

Table 14
DeLong test results.

Model 1	Model 2	ΔROC-AUC	p-value	Significant ($p < 0.05$
Multimodal	ViT-b-16	0.16593	5.25794×10^{-31}	Yes
Multimodal	Swin-b	0.27571	1.74715×10^{-54}	Yes
Multimodal	Swin-V2-b	0.21382	2.33602×10^{-37}	Yes
Multimodal	1D-CNN	0.15276	1.46258×10^{-30}	Yes
ViT-b-16	Swin-b	0.10978	3.30458×10^{-13}	Yes
ViT-b-16	Swin-V2-b	0.04790	4×10^{-5}	Yes
ViT-b-16	1D-CNN	−0.01317	4.7522×10^{-1}	No
Swin-b	Swin-V2-b	−0.06188	2.30267×10^{-6}	Yes
Swin-b	1D-CNN	−0.12295	1.73326×10^{-8}	Yes
Swin-V2-b	1D-CNN	−0.06106	2.91×10^{-3}	Yes

and Model 2 is wrong, and c represents the opposite. A low p-value ($p < 0.05$), allows rejection of the null hypothesis (H_0), confirming that one model performs substantially better. Results show the multimodal model significantly outperforms all unimodal models, whereas some unimodal pairs such as ViT-b-16 and the 1D-CNN do not show significant differences. This highlights that numeric biomarkers alone capture substantial discriminative information, performing comparably to complex transformer architectures.

Narrow CI ranges of the multimodal model indicate stable, consistent performance, whereas unimodal models show wider intervals, reflecting higher variance. DeLong's test in Table 14 confirms statistically significant improvements in ROC-AUC for the multimodal model over baseline models. McNemar's test in Table 15 further demonstrates that the multimodal model significantly outperforms competing models in classification decisions, whereas some unimodal models show no significant difference. These results collectively support the claim that the proposed model achieves robust performance gains under rigorous statistical validation.

4.4. Deployment feasibility

Model deployment bridges the gap between model development and real-world clinical application. Evaluating deployment feasibility is therefore essential for validating both the effectiveness and practical usability of the proposed framework. One of the key challenges in deploying deep learning models is balancing predictive accuracy with rapid inference and computational efficiency. Table 16 reports the inference times for the complete glaucoma classification pipeline and demonstrates that the proposed system can be deployed without compromising performance.

To improve efficiency, the pipeline is designed as a modular system rather than performing a single computationally expensive end-to-end prediction. The workflow decomposes the task into segmentation and measurement stages that first extract clinically relevant glaucoma biomarkers from retinal fundus images. These extracted features are then used for multimodal classification. This design ensures that computationally intensive vision transformer models are used only during the final prediction stage, while earlier pipeline stages rely on lighter models for biomarker extraction. As a result, most of the processing workload is handled by lightweight components, limiting the computational burden of transformer-based models.

The complete end-to-end pipeline requires approximately 2.85 s per image on a CPU and 0.30 s on a GPU, demonstrating that the proposed framework can operate efficiently even in environments with limited hardware resources. Inference times were evaluated using an Intel(R) Xeon(R) CPU (2.20 GHz, 32 GB RAM) and a Tesla P100-PCIE-16GB GPU, with the GPU achieving nearly a 10× speedup for multimodal classification and approximately a 9.5× speedup for the overall pipeline.

Shyamalee et al. [61] develop and deploy a Streamlit powered application (GlaucoCare) for glaucoma classification from retinal fundus images. For practical validation, the system was deployed via a Gradio web interface hosted on Hugging Face Spaces, enabling accessible, reproducible evaluation of the framework. Since CPU inference times were within acceptable limits, the final deployment utilized the default Hugging Face CPU environment, eliminating the need for specialized hardware. These results demonstrate that the proposed pipeline effectively balances predictive performance, computational efficiency, and scalability, supporting its feasibility for real-world glaucoma screening applications.

4.5. Model interpretability

Model interpretability is an essential aspect of deep learning as it enables researchers, clinicians and medical practitioners to understand the factors influencing model predictions, while also supporting fairness and trust in clinical deployment. We hypothesize that this cross-modal interaction retrieves clinically relevant biomarkers, as supported by interpretability analyses that show a strong focus on the optic disc and cup regions via cross-attention maps and Grad-CAM visualizations. Additionally, the t-SNE visualization of the fused feature space in Fig. 11 achieves a KL divergence of 0.23726 after 1000 iterations, demonstrating clear class separation. For the image modality, we employ explainable AI (xAI) techniques such as Grad-CAM [73] and Integrated Gradients (IG) [74] to identify regions that significantly influence model predictions. Grad-CAM is employed to generate heatmaps for the Swin-based vision transformers, highlighting the areas of the fundus images that contribute the most to the classification decision. Complementarily, IG assigns importance scores to the individual pixels, providing a finer grained attribution of model behavior. Both methods consistently depict regions around the optic disc and optic cup, suggesting that the structural characteristics of these regions play a key role in glaucoma classification. The regions which significantly drive model predictions are shown in Figs. 12 and 13.

Table 15
McNemar test results.

Model 1	Model 2	b	c	McNemar statistic	p-value	Significant ($p < 0.05$)
Multimodal	ViT-b-16	194	35	110.39738	8.01910×10^{-26}	Yes
Multimodal	Swin-b	271	39	173.62581	1.19482×10^{-39}	Yes
Multimodal	Swin-V2-b	221	41	123.66412	9.97819×10^{-29}	Yes
Multimodal	1D-CNN	176	25	113.43781	1.73024×10^{-26}	Yes
ViT-b-16	Swin-b	120	47	31.91018	1.61469×10^{-8}	Yes
ViT-b-16	Swin-V2-b	61	40	4.36634	3.666×10^{-2}	Yes
ViT-b-16	1D-CNN	134	142	0.23188	6.3013×10^{-1}	No
Swin-b	Swin-V2-b	46	98	18.77778	1×10^{-5}	Yes
Swin-b	1D-CNN	127	208	19.58507	9.62179×10^{-6}	Yes
Swin-V2-b	1D-CNN	140	169	2.72168	9.899×10^{-2}	No

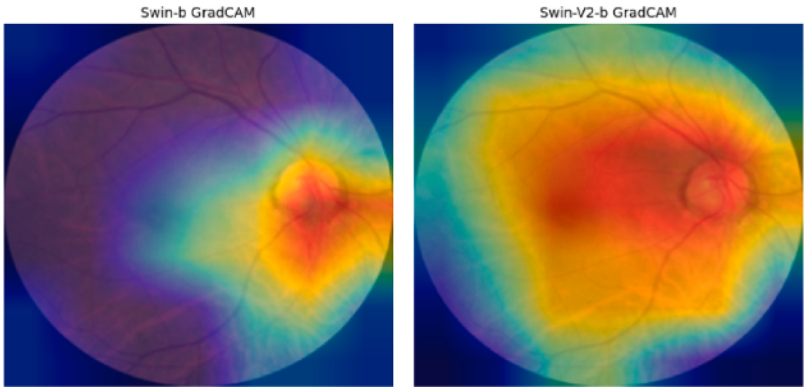


Fig. 12. Grad-CAM for Swin-b and Swin-V2-b.

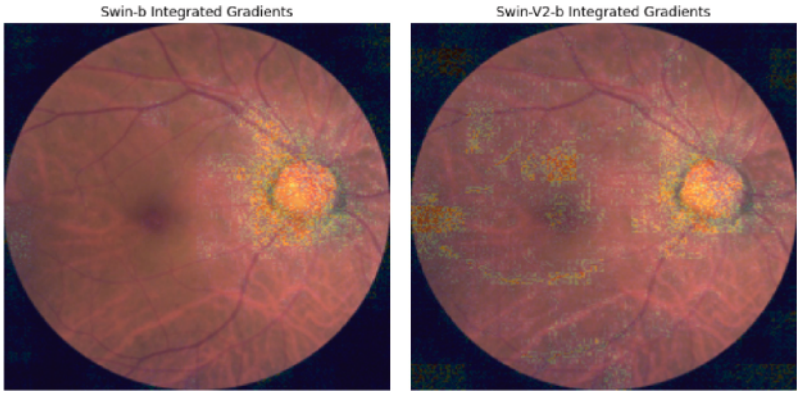


Fig. 13. Integrated Gradients for Swin-b and Swin-V2-b.

In addition to image based interpretability, we analyze the contributions of the numeric modality, which consists of clinically derived biomarkers. Feature attribution scores are computed to quantify the influence of each feature on the final prediction, where positive attributions increase the likelihood of glaucoma classification and negative attributions decrease it. Fig. 14 depicts the top 15 most significant features contributing to the predictions made by the model. Attributions are based on the input features and quantify how the individual representations affect the final prediction. The results indicate that biomarkers related to the optic cup such as the radius, area and circumference are among the most influential features. This finding is consistent with clinical literature [75–77] and the regions in the images that the vision transformers attend to the most as shown in Figs. 12 and 13. Other features, including the retina circumference, macula circumference and vasculature density also contribute to the model's predictions, aligning with known clinical indicators [75,76]. This provides further credence to our claims that numerically extracted biomarkers influence glaucoma prediction outcomes.

To better understand the interactions between modalities, we examine feature activations before and after the application of cross-modal attention in Fig. 15. These feature activations help us understand how inter-modal interactions contribute to the refinement of modality specific representations. We observe that the magnitude of the image based feature activations decreased after cross attention, while the magnitude of the numeric activations increased. This suggests that the model increased its focus on the numeric biomarkers after the introduction of cross attention.

To quantify this interaction, we compute the L_2 displacement between feature representations before and after cross attention, as well as the cosine similarity between the corresponding feature vectors. The L_2 displacement measures the magnitude of change, while the cosine similarity captures the directional shift in the feature space. For feature vector $\tau \in \mathbb{R}^d$, where $d = 256$, we have:

$$L_2 = \|\tau_{\text{after}} - \tau_{\text{before}}\|_2 = \sqrt{\sum_{i=1}^{d} (\tau_{\text{after}}^{(i)} - \tau_{\text{before}}^{(i)})^2} \tag{1}$$

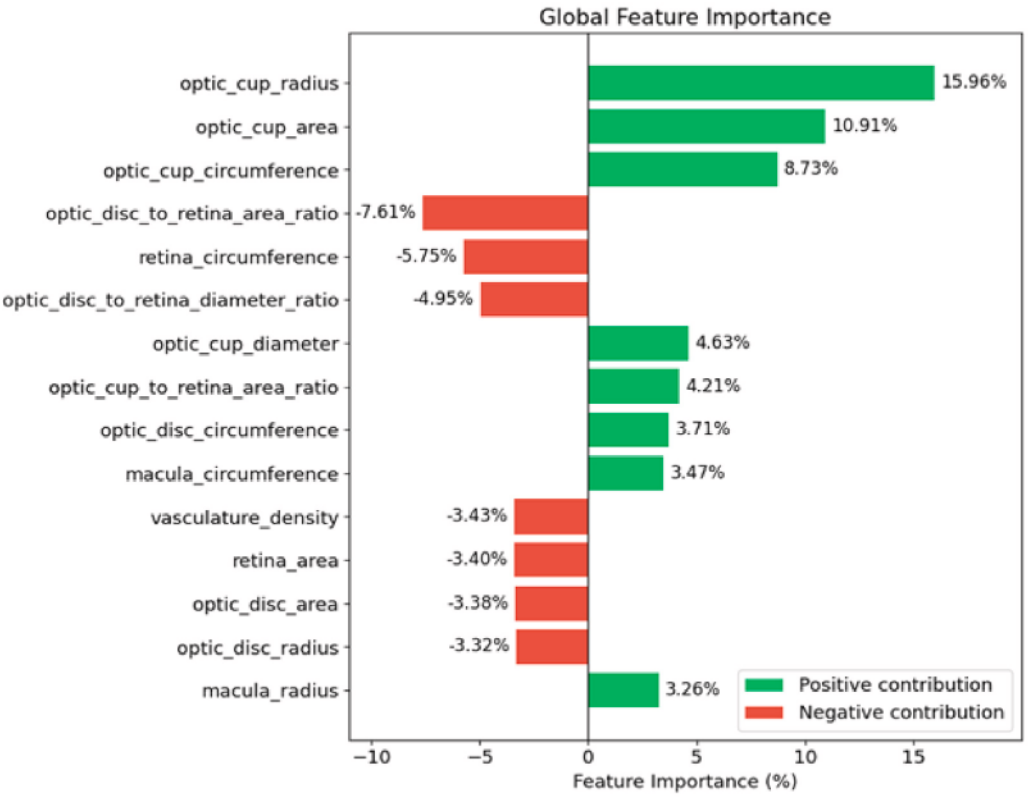


Fig. 14. Top 15 global feature attributions for the 1D-CNN.

Table 16
Inference times for assessing deployment feasibility.

Pipeline component	CPU (time in seconds)	GPU (time in seconds)
Optic Disc Diameter	0.20626 ± 0.00373	0.00436 ± 0.00005
Optic Cup Diameter	0.20836 ± 0.00287	0.00488 ± 0.00015
Retinal Vasculature Density	0.20868 ± 0.00358	0.00950 ± 0.00040
Macula Area	0.69870 ± 0.04735	0.11352 ± 0.01661
Retinal Diameter	0.01520 ± 0.00020	0.02276 ± 0.01948
Multimodal Classification	1.08506 ± 0.01679	0.12304 ± 0.00171
Total Time	2.42226 ± 0.07452	0.27806 ± 0.0384

In Equation (1), τ_{before} denotes the output of the layers before cross attention, and τ_{after} represents the output after cross attention. An L_2 value near zero implies that the modalities ignored each other, moreover, a θ value of 0° ($cos(0°) = 1$), would signify that the other modality has had no noticeable influence on the directional representation of the current modality. From Fig. 16 the L_2 magnitude is observed to be 12.53 and 13.88 for the numeric and image modality, respectively. For the numeric modality, $\theta = 71.2°$ ($cos(71.2°) = 0.32227$) indicates that the image modality pulled the numeric feature representations towards a new region of the feature space. Similarly, the numeric modality was able to pull the image representations by an angle of $\theta = 64.8°$ ($cos(64.8°) = 0.42578$). The observed values indicate that both modalities undergo substantial transformations, with notable angular deviations, suggesting that cross-modal attention introduces meaningful changes to the feature representations.

We further analyze the query-key similarity using a scaled dot product attention score for the cross attention blocks, to understand how modalities attend to each. The absolute magnitude of similarity between image queries and numeric keys was found to be 1.1618, whereas that for the numeric queries and image keys was found to be 0.5833. Fig. 17 highlights the numeric keys that the images attend to using cross modal attention. We observe that the images heavily focus on the retina circumference, retina radius, and the optic cup circumference. Other important biomarkers with an absolute gradient magnitude more than 2 include the optic cup and disc diameters, the vasculature density, and other glaucoma biomarkers based on the optic cup and disc. This is consistent with our hypothesis that the image representations would retrieve clinically relevant biomarkers. It is also consistent with data used in practice to inform glaucoma classification [75]. The Grad-CAM visualization in Fig. 12 highlights the area around the optic disc and cup as the most significant. We also observe this phenomenon while analyzing which numeric features the image representations attend to. This provides proof that the model is learning to capture both intra and inter-modality feature representations.

Additionally, we investigate which image embedding dimensions are most influenced by the numeric modality. Fig. 18 shows the top 10 image dimensions that were attended to most by the numeric features. The results highlight that the numeric features attend selectively to a subset of image embedding dimensions, indicating targeted cross-modal interaction rather than uniform feature mixing. This suggests that numeric biomarkers guide the model towards picking specific visual representations that are relevant for glaucoma detection.

Finally, we validate the importance of the learned features using RemOve And Retrain (ROAR) [78] experiments. Considering a set of all features learnt by the model (ϕ_l), ordered in terms of their relative importance, we iteratively remove the top k% of features and observe the degradation in performance of the model. Here, $k \in [0, 25, 50]$ and is systematically masked from the feature set ϕ_l. The modified masked feature set denoted by ϕ_{m_k} is used to retrain the model. If the removal of

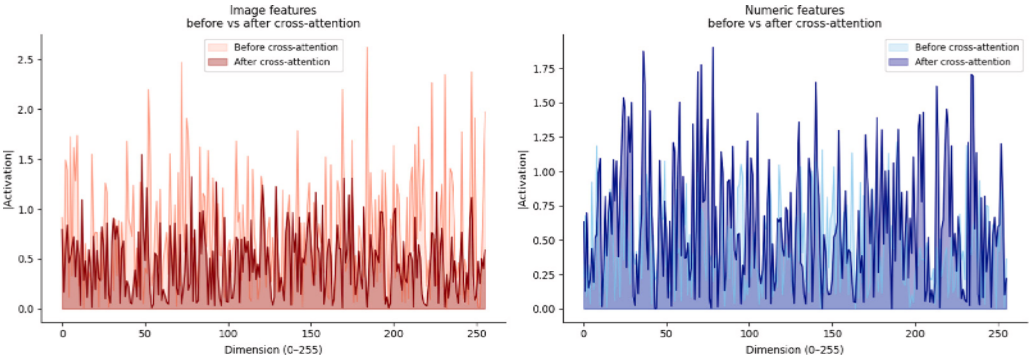


Fig. 15. Feature activations for the image and numeric modality before and after cross-attention.

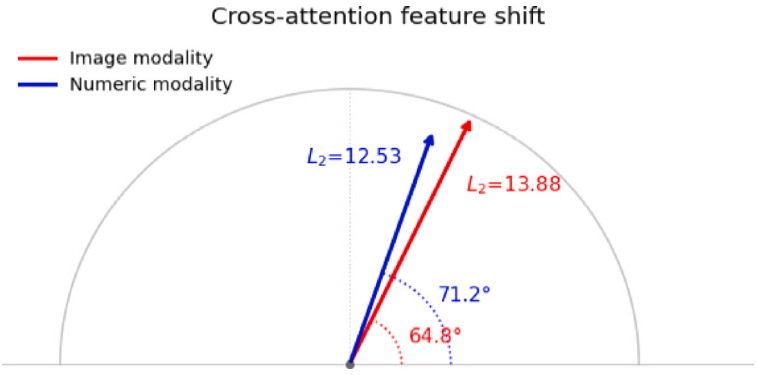


Fig. 16. Cross-attention based feature shift.

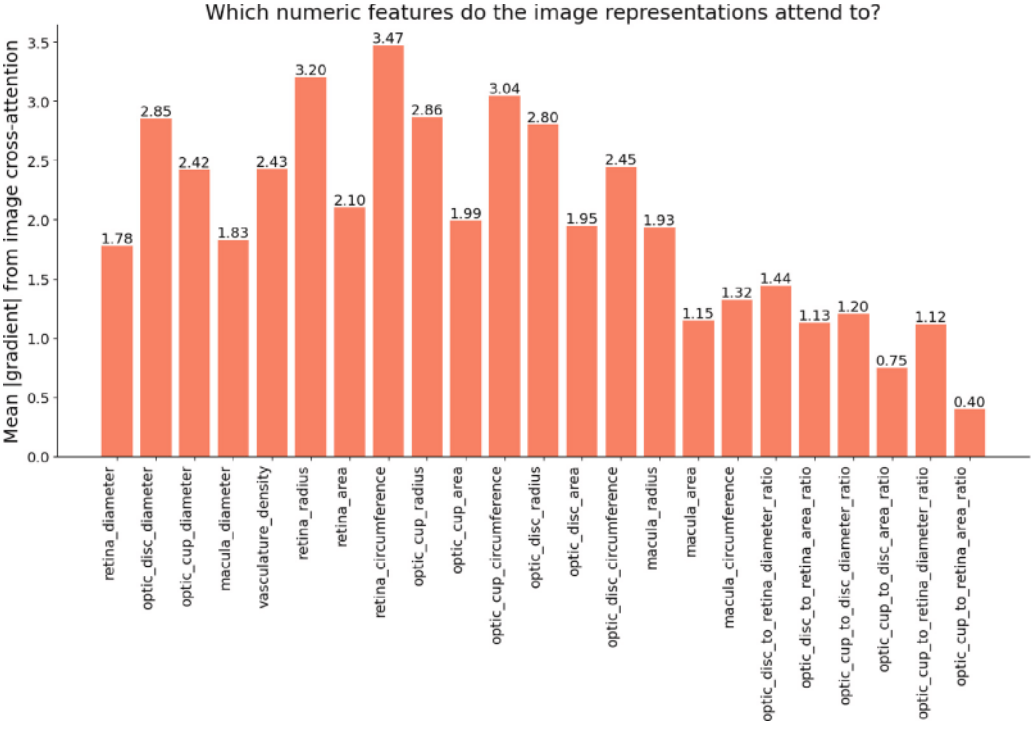


Fig. 17. Numeric keys that images attend to for glaucoma classification.

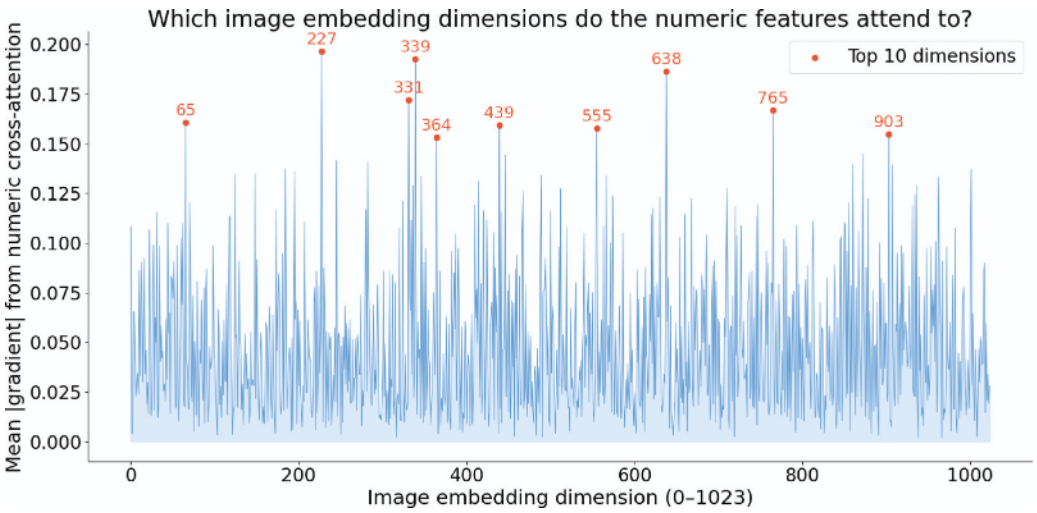


Fig. 18. Image Embeddings that Numeric Features Focus On.

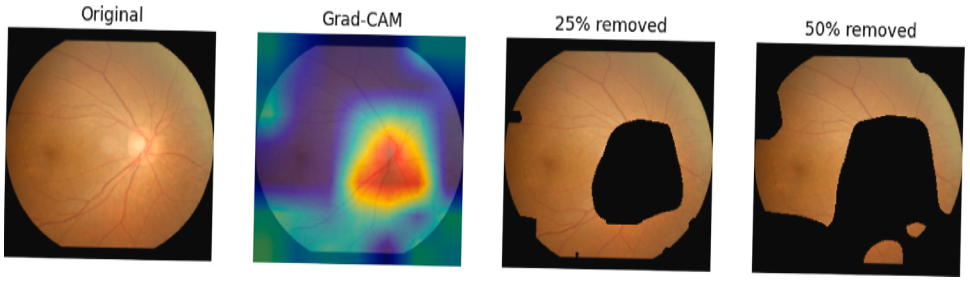


Fig. 19. Masking the top 25% and 50% of Grad-CAM features.

Table 17
ROAR results after masking the top $k\%$ of features.

Masking values (Top $k\%$)	Test accuracy	Test F_1-Score
0	0.92630	0.91677
25	0.56349	0.66258
50	0.45351	0.45352

the top $k\%$ of features causes a significant reduction in the performance of the model, it implies that the features learnt were vital for correct glaucoma classification. Fig. 19 shows the masking of the top 25% and 50% of images features. Similarly, the top $k\%$ of numeric features are also masked. The model is retrained on the masked dataset and the results are reported in Table 17. Masking the top 25% of features leads to a significant decline in performance, indicating that these features, especially the optic disc, optic cup, and associated biomarkers (e.g., disc and cup measurements along with retinal geometry) are essential for glaucoma classification. In their absence, the retrained model fails to learn sufficiently discriminative representations for accurate prediction. On increasing the masking to 50% the model cannot adequately learn the vasculature density of the fundus images, leading to a further drop in performance as expected. These results confirm that the model relies on meaningful and informative features for glaucoma prediction.

5. Conclusion

The proposed study contributes to glaucoma research by developing a multimodal framework that integrates segmentation derived clinical biomarkers with deep learning derived image features to improve glaucoma classification. The primary focus of this work is the extraction of clinically meaningful anatomical measurements from retinal fundus images and their integration with transformer-based visual features to enhance diagnostic performance.

Segmentation models play a central role in the proposed pipeline. A U-Net based segmentation model is used to identify variations in optic cup and optic disc structures, enabling the extraction of important biomarkers such as optic disc area and optic cup-to-disc ratio. In addition, a SegFormer based macula segmentation model provides complementary anatomical information, further improving classification performance. Using these segmentation outputs, several geometric features, including radius, diameter, area, and circumference are derived through feature engineering techniques. Analysis of these biomarkers shows a consistent increase in the optic cup-to-disc ratio among glaucoma patients, consistent with established clinical observations thereby, validating the effectiveness of the segmentation based feature extraction process.

The proposed multimodal architecture integrates two complementary processing branches. The image branch uses vision transformers to capture global contextual information from fundus images via attention mechanisms such as multi-head attention (MHA), window-based multi-head self-attention (W-MSA), and shifted-window multi-head self-attention (SW-MSA). The numeric branch processes the extracted glaucoma biomarkers using a 1D convolutional neural network (1D-CNN), enabling efficient learning from structured numerical features. In the final stage, a weighted fusion of image-based features and clinical biomarkers is used to classify glaucoma.

The proposed framework achieves an ROC-AUC score of 0.97464 and an accuracy of 92.63%, demonstrating strong performance in glaucoma detection. By combining segmentation derived biomarkers with transformer based image representations, the model provides a scalable

approach for automated glaucoma screening from retinal fundus images. These results highlight the potential of multimodal deep learning methods for improving the reliability of computer aided diagnosis in ophthalmology.

6. Future work

The multimodal framework developed in this study demonstrates strong potential for improving automated glaucoma classification. Future work will focus on extending the proposed approach by incorporating additional imaging modalities, particularly optical coherence tomography (OCT), to further enhance the availability of clinically relevant structural information.

Another important direction is to explore different combinations of vision transformer architectures to better understand their contributions to multimodal learning and identify configurations that achieve improved performance and computational efficiency. In addition, optimizing the model for real-world deployment will be a key objective. This includes developing lightweight implementations that maintain high classification accuracy while enabling faster inference and efficient on-device execution. Achieving an optimal balance between predictive performance and computational efficiency will be essential for practical clinical deployment of multimodal glaucoma screening systems.

CRediT authorship contribution statement

Rahil Parikh: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Van Nguyen:** Writing – review & editing, Validation, Investigation. **Anita Penkova:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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